

Seedrun v2: small-M sweep with bias-free ortho and B_full slicing

Sweep: M in {3, 4, 5, 8, 9} x $P \leq M$ x 3 seeds x {vanilla, ortho}

Total: 114 runs ($M=3:2$, $M=4:3$, $M=5:4$, $M=8:5$, $M=9:5$ P-pairs).

Epochs: 1000 alpha: 1.0

B: one fixed master B in $R^{\{3 \times 20\}}$ (seed_B = 42); each
run uses $B[:, :M]$ as its frozen B .

Ortho: no bias inside ReLU (pure rotation Qz).

Headline (D_H_state, state-space Hellinger)

pooled mean rel gap	: -4.93%
bootstrap 95% CI	: [-9.11%, -1.35%]
ortho wins on mean	: 14/19
sign-test p (one-sided)	: 0.0318
Wilcoxon p (one-sided)	: 0.0162

Headline (PSE, power-spectrum Hellinger)

pooled mean rel gap	: +nan%
bootstrap 95% CI	: [+nan%, +nan%]
ortho wins on mean	: 12/19
sign-test p	: 0.1796

Statistics: PSE (Hellinger of power spectrum)

Per-cell paired statistics on PSE (Hellinger of power spectrum)

M	P	vanilla	ortho	rel_gap%	d_z	95% t-CI on diff	wilc p	wins
3	2	0.3712±0.001	0.3624±0.118	-2.4%	-0.07	[-0.3012, +0.2836]	0.625	2/3
3	3	0.2740±0.000	nan± nan	+nan%	+nan	[+nan, +nan]	nan	0/3
4	2	0.2421±0.002	0.3014±0.111	+24.5%	+0.53	[-0.2209, +0.3394]	0.875	1/3
4	3	0.2987±0.111	0.1582±0.092	-47.0%	-0.82	[-0.5672, +0.2863]	0.250	2/3
4	4	0.3678±0.016	0.4702±0.099	+27.9%	+1.08	[-0.1327, +0.3377]	1.000	0/3
5	2	0.4409±0.070	0.4170±0.086	-5.4%	-0.24	[-0.276, +0.2281]	0.375	2/3
5	3	0.4091±0.104	0.1653±0.081	-59.6%	-1.59	[-0.6245, +0.1369]	0.125	3/3
5	4	0.3059±0.154	0.1800±0.172	-41.1%	-0.39	[-0.9246, +0.6729]	0.250	2/3
5	5	0.3889±0.006	0.3101±0.198	-20.3%	-0.41	[-0.5572, +0.3997]	0.375	2/3
8	2	0.2548±0.163	0.3339±0.041	+31.1%	+0.44	[-0.3692, +0.5275]	0.625	2/3
8	3	0.2438±0.165	0.2120±0.021	-13.1%	-0.19	[-0.4409, +0.3772]	0.625	2/3
8	4	0.1158±0.032	0.0833±0.005	-28.1%	-0.93	[-0.1197, +0.05469]	0.125	3/3
8	5	0.0800±0.006	0.0864±0.004	+8.0%	+0.62	[-0.01921, +0.03202]	0.875	1/3
8	6	0.1810±0.089	0.0688±0.022	-62.0%	-1.08	[-0.3704, +0.146]	0.250	2/3
9	2	0.3532±0.049	0.3653±0.032	+3.4%	+0.51	[-0.04679, +0.07096]	0.875	1/3
9	3	0.3270±0.015	0.1855±0.126	-43.3%	-1.14	[-0.451, +0.1681]	0.250	2/3
9	4	0.1846±0.159	0.0699±0.027	-62.1%	-0.63	[-0.5694, +0.34]	0.125	3/3
9	5	0.0867±0.001	0.0770±0.011	-11.1%	-0.94	[-0.03504, +0.01576]	0.250	2/3
9	6	0.0718±0.007	0.0724±0.007	+0.9%	+0.05	[-0.02902, +0.03026]	0.625	2/3

Pooled across cells

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mean rel gap      : +nan%
bootstrap 95% CI  : [+nan%, +nan%]
wins              : 12/19
sign-test p       : 0.1796
Wilcoxon p        : nan

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Stratified by M

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M=3  mean rel gap =  +nan%   (1/2)
M=4  mean rel gap =  +1.77%  (1/3)
M=5  mean rel gap = -31.61%  (4/4)
M=8  mean rel gap = -12.80%  (3/5)
M=9  mean rel gap = -22.45%  (3/5)

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Stratified by P

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P=2  mean rel gap = +10.23%  (2/5)
P=3  mean rel gap =  +nan%   (4/5)
P=4  mean rel gap = -25.87%  (3/4)
P=5  mean rel gap =  -7.79%  (2/3)
P=6  mean rel gap = -30.56%  (1/2)

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Statistics: D_H state-space (Hellinger)

Per-cell paired statistics on D_H state-space (Hellinger)

M	P	vanilla	ortho	rel_gap%	d_z	95% t-CI on diff	wilc p	wins
3	2	0.9476±0.000	0.9778±0.010	+3.2%	+3.01	[+0.005294, +0.05519]	1.000	0/3
3	3	0.9906±0.000	0.9590±0.040	-3.2%	-0.79	[-0.1308, +0.06774]	0.125	3/3
4	2	0.9884±0.000	0.9360±0.042	-5.3%	-1.23	[-0.1582, +0.05338]	0.125	3/3
4	3	0.9430±0.045	0.8830±0.076	-6.4%	-1.48	[-0.1608, +0.04072]	0.125	3/3
4	4	0.9305±0.013	0.9162±0.059	-1.5%	-0.30	[-0.1343, +0.1057]	0.375	2/3
5	2	0.9364±0.066	0.9549±0.007	+2.0%	+0.28	[-0.1441, +0.1812]	0.625	2/3
5	3	0.9362±0.059	0.8048±0.136	-14.0%	-1.66	[-0.3283, +0.06542]	0.125	3/3
5	4	0.9683±0.036	0.7977±0.156	-17.6%	-1.09	[-0.5605, +0.2192]	0.125	3/3
5	5	0.9071±0.037	0.8840±0.011	-2.5%	-0.69	[-0.1057, +0.0595]	0.250	2/3
8	2	0.9504±0.022	0.9571±0.016	+0.7%	+0.42	[-0.03272, +0.04603]	0.875	1/3
8	3	0.8453±0.124	0.8876±0.023	+5.0%	+0.36	[-0.2487, +0.3335]	0.750	1/3
8	4	0.7859±0.086	0.7614±0.088	-3.1%	-1.09	[-0.08034, +0.03145]	0.125	3/3
8	5	0.6462±0.039	0.6455±0.038	-0.1%	-0.05	[-0.03283, +0.03147]	0.625	2/3
8	6	0.8343±0.045	0.6093±0.059	-27.0%	-2.20	[-0.4795, +0.02938]	0.125	3/3
9	2	0.9431±0.028	0.9192±0.055	-2.5%	-0.81	[-0.09714, +0.04947]	0.250	2/3
9	3	0.9450±0.024	0.9211±0.034	-2.5%	-2.03	[-0.05301, +0.005338]	0.125	3/3
9	4	0.8005±0.076	0.6247±0.149	-22.0%	-1.03	[-0.6001, +0.2485]	0.125	3/3
9	5	0.6496±0.006	0.6347±0.024	-2.3%	-0.73	[-0.06603, +0.03616]	0.250	2/3
9	6	0.5827±0.038	0.6146±0.054	+5.5%	+0.35	[-0.196, +0.2597]	0.750	1/3

Pooled across cells

```

mean rel gap      : -4.93%
bootstrap 95% CI  : [-9.11%, -1.35%]
wins              : 14/19
sign-test p       : 0.0318
Wilcoxon p        : 0.0162

```

Stratified by M

```

M=3  mean rel gap = +0.00%  (1/2)
M=4  mean rel gap = -4.40%  (3/3)
M=5  mean rel gap = -8.06%  (3/4)
M=8  mean rel gap = -4.90%  (3/5)
M=9  mean rel gap = -4.77%  (4/5)

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Stratified by P

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P=2  mean rel gap = -0.39%  (2/5)
P=3  mean rel gap = -4.22%  (4/5)
P=4  mean rel gap = -11.06% (4/4)
P=5  mean rel gap = -1.65%  (3/3)
P=6  mean rel gap = -10.75% (1/2)

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Can ortho at smaller M match vanilla at bigger M? (v2)

Central question (v2): does ortho at smaller M match vanilla at bigger M? Means over 3 seeds; ratio = ortho_small / vanilla_big.
ratio <= 0.95 -> ortho strictly better
0.95 < r <=1.05 -> match
r > 1.05 -> vanilla wins

PSE (Hellinger of power spectrum)

P	M_small	best M_big	ortho mean	vanilla mean	ratio	verdict
2	M=3	M=9	0.3624	0.3532	1.026	match
3	M=3	(none)	0.4404	-	-	no impr
2	M=4	M=9	0.3014	0.3532	0.853	ortho >>
3	M=4	M=9	0.1582	0.3270	0.484	ortho >>
4	M=4	(none)	0.4702	-	-	no impr
2	M=5	(none)	0.4170	-	-	no impr
3	M=5	M=9	0.1653	0.3270	0.505	ortho >>
4	M=5	M=9	0.1800	0.1846	0.975	match
5	M=5	(none)	0.3101	-	-	no impr
2	M=8	M=9	0.3339	0.3532	0.945	ortho >>
3	M=8	M=9	0.2120	0.3270	0.648	ortho >>
4	M=8	M=9	0.0833	0.1846	0.451	ortho >>
5	M=8	M=9	0.0864	0.0867	0.996	match
6	M=8	M=9	0.0688	0.0718	0.959	match

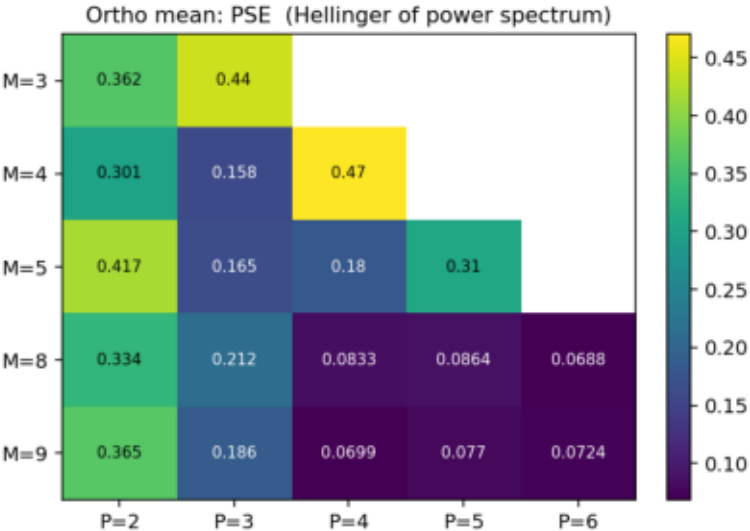
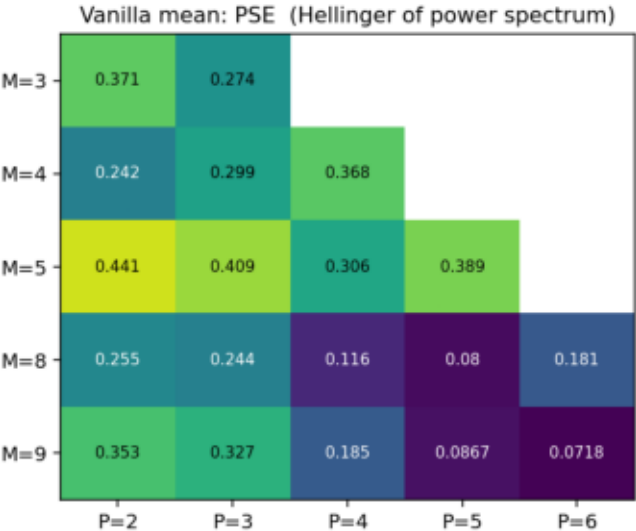
free-M-reduction count: 10/14 (71%)

D_H state-space (Hellinger)

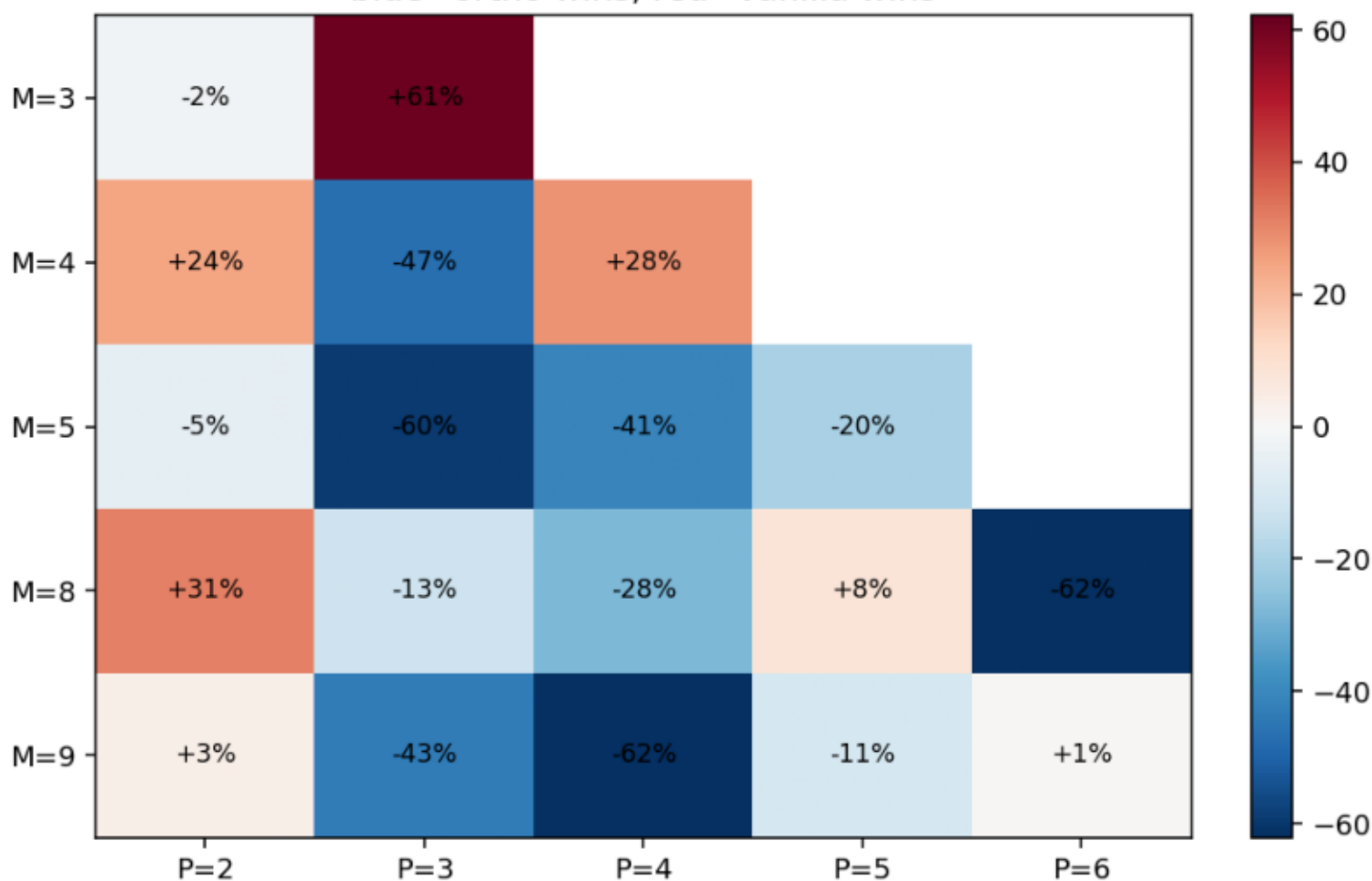
P	M_small	best M_big	ortho mean	vanilla mean	ratio	verdict
2	M=3	M=9	0.9778	0.9431	1.037	match
3	M=3	M=9	0.9590	0.9450	1.015	match
2	M=4	M=9	0.9360	0.9431	0.992	match
3	M=4	M=9	0.8830	0.9450	0.934	ortho >>
4	M=4	M=5	0.9162	0.9683	0.946	ortho >>
2	M=5	M=9	0.9549	0.9431	1.013	match
3	M=5	M=9	0.8048	0.9450	0.852	ortho >>
4	M=5	M=9	0.7977	0.8005	0.996	match
5	M=5	(none)	0.8840	-	-	no impr
2	M=8	M=9	0.9571	0.9431	1.015	match
3	M=8	M=9	0.8876	0.9450	0.939	ortho >>
4	M=8	M=9	0.7614	0.8005	0.951	match
5	M=8	M=9	0.6455	0.6496	0.994	match
6	M=8	M=9	0.6093	0.5827	1.046	match

free-M-reduction count: 13/14 (93%)

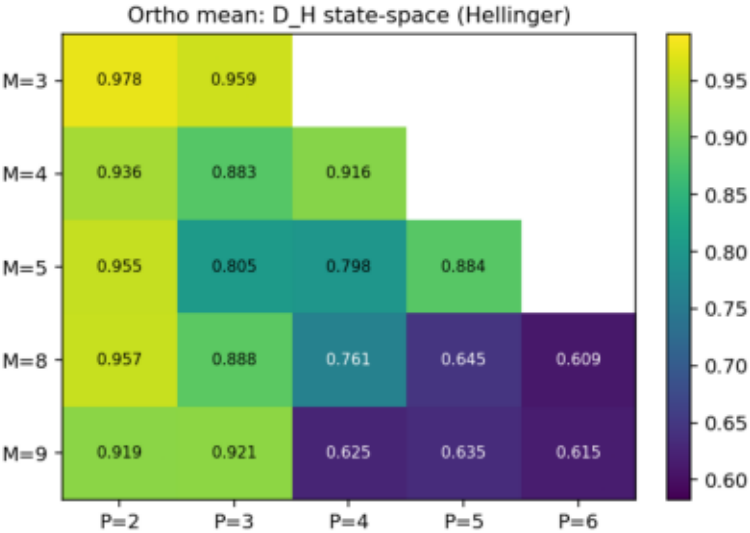
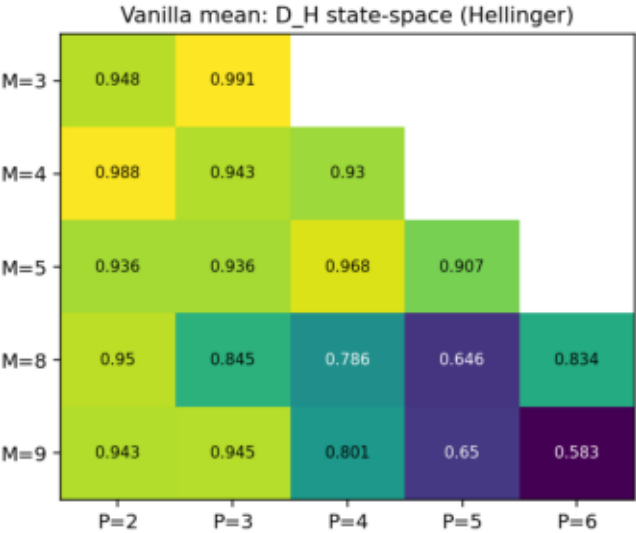
PSE (Hellinger of power spectrum) (mean over 3 seeds, lower better)



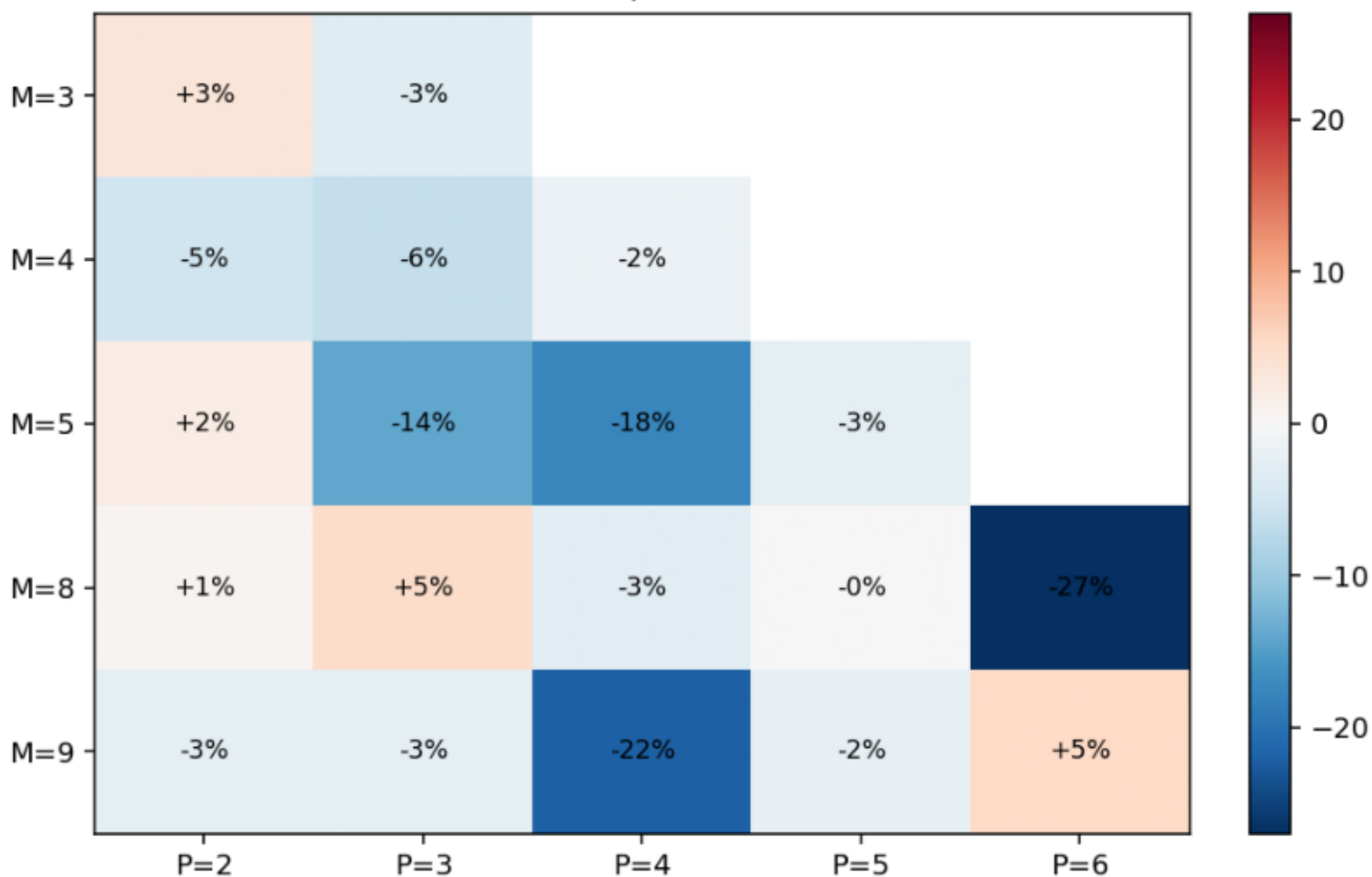
Gap (ortho-vanilla)/vanilla on mean PSE (Hellinger of power spectrum)
blue=ortho wins, red=vanilla wins



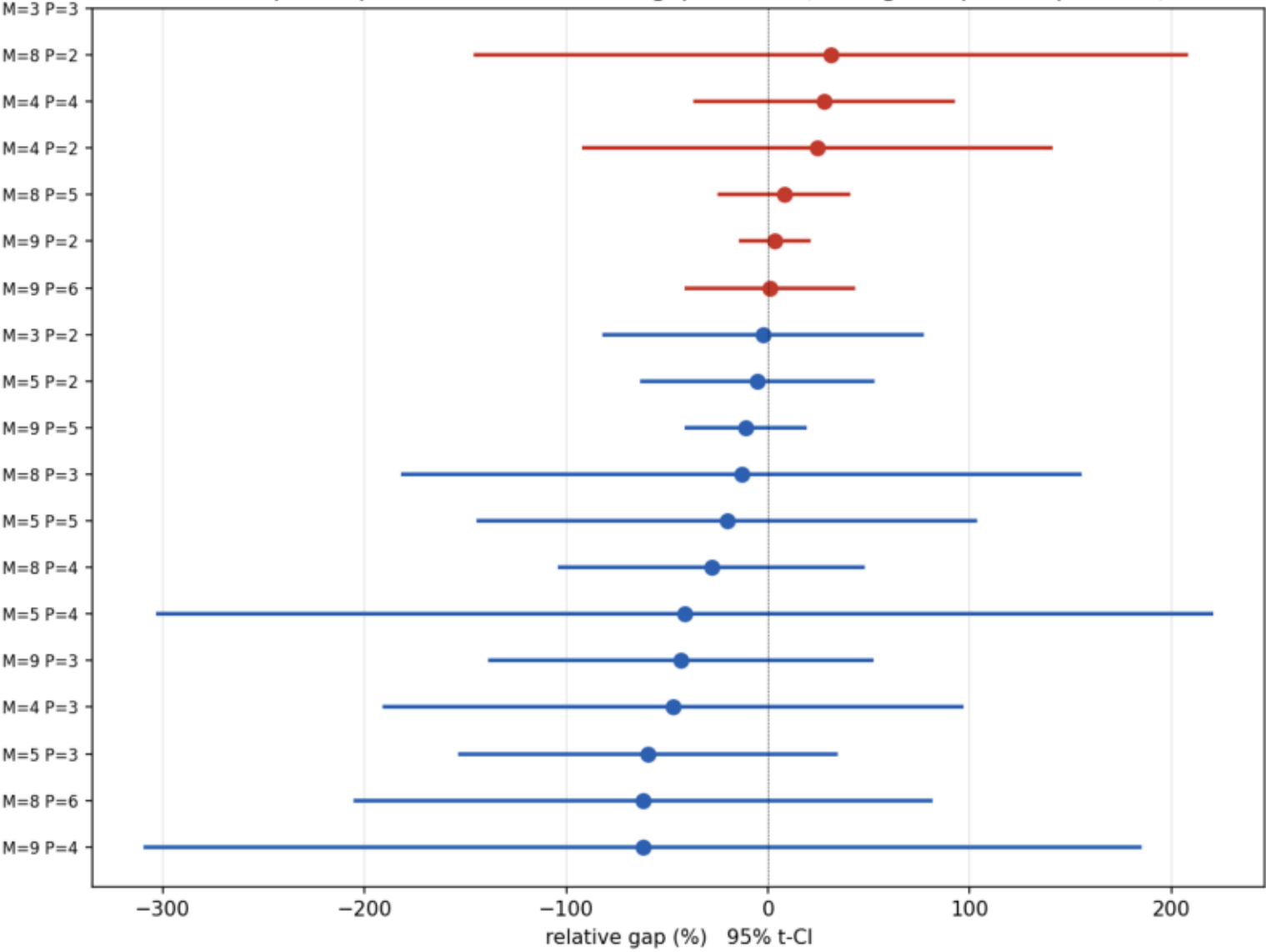
D_H state-space (Hellinger) (mean over 3 seeds, lower better)



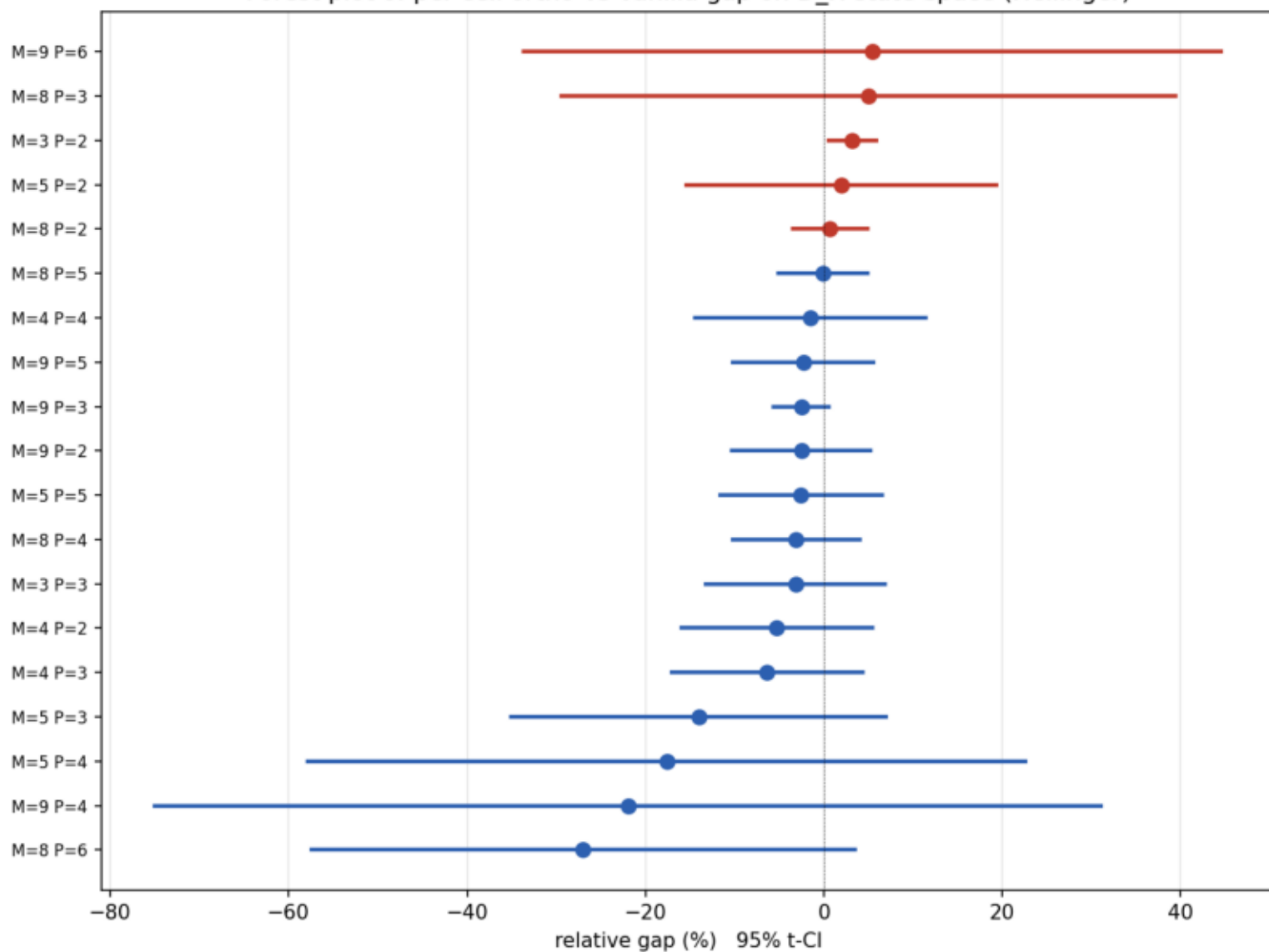
Gap (ortho-vanilla)/vanilla on mean D_H state-space (Hellinger)
blue=ortho wins, red=vanilla wins



Forest plot of per-cell ortho-vs-vanilla gap on PSE (Hellinger of power spectrum)



Forest plot of per-cell ortho-vs-vanilla gap on D_H state-space (Hellinger)



Highlight trajectories

Each of the next pages shows one (M, P) cell at the median-PSE seed: 3-D views (top row) and x-z 2-D projections (bottom row) for ground truth / vanilla / orthogonal.

Order: best ortho wins first on D_H_state (state-space Hellinger).

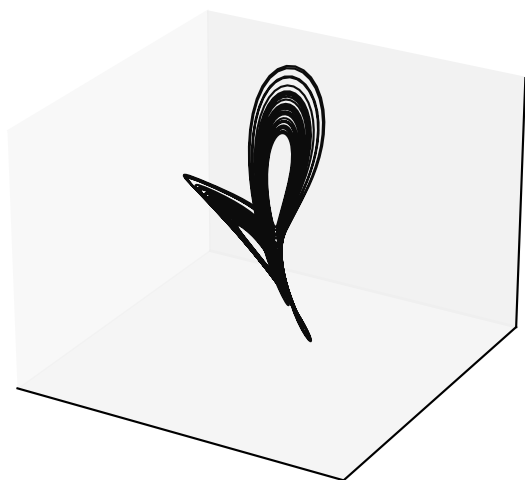
M=8	P=6	D_H gap =	-27.0%	PSE gap =	-62.0%
M=9	P=4	D_H gap =	-22.0%	PSE gap =	-62.1%
M=5	P=4	D_H gap =	-17.6%	PSE gap =	-41.1%
M=5	P=3	D_H gap =	-14.0%	PSE gap =	-59.6%
M=4	P=3	D_H gap =	-6.4%	PSE gap =	-47.0%
M=4	P=2	D_H gap =	-5.3%	PSE gap =	+24.5%

Failure cell (worst ortho regression):

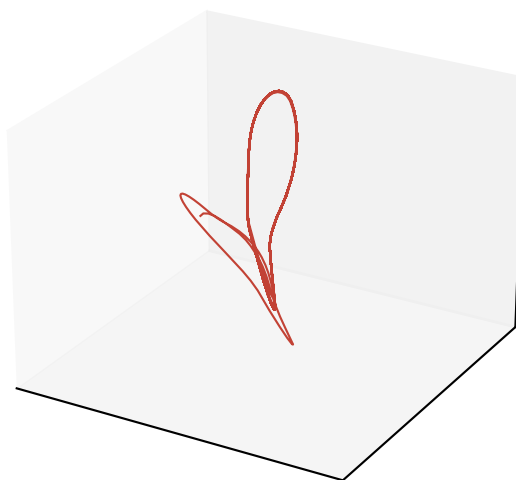
M=9	P=6	D_H gap =	+5.5%
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M=8 P=6 (median-PSE seed=1) PSE: van 0.181 ± 0.089 ortho 0.069 ± 0.022 (-62.0%) D_H: van 0.834 ± 0.045 ortho 0.609 ± 0.059 (-27.0%)

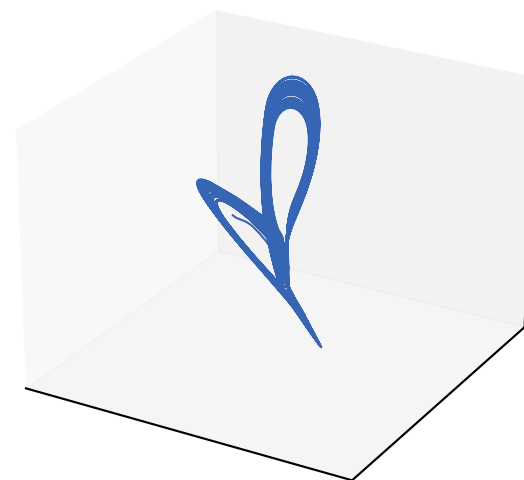
Truth



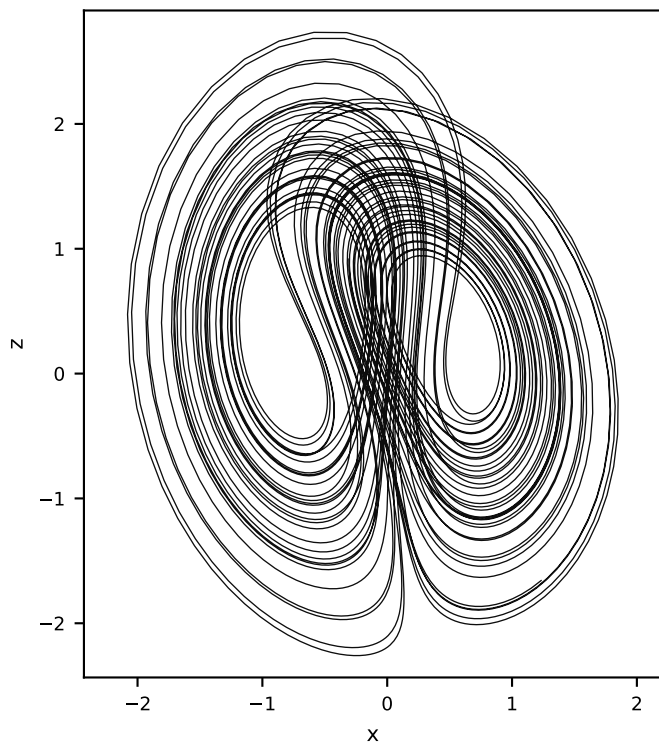
Vanilla



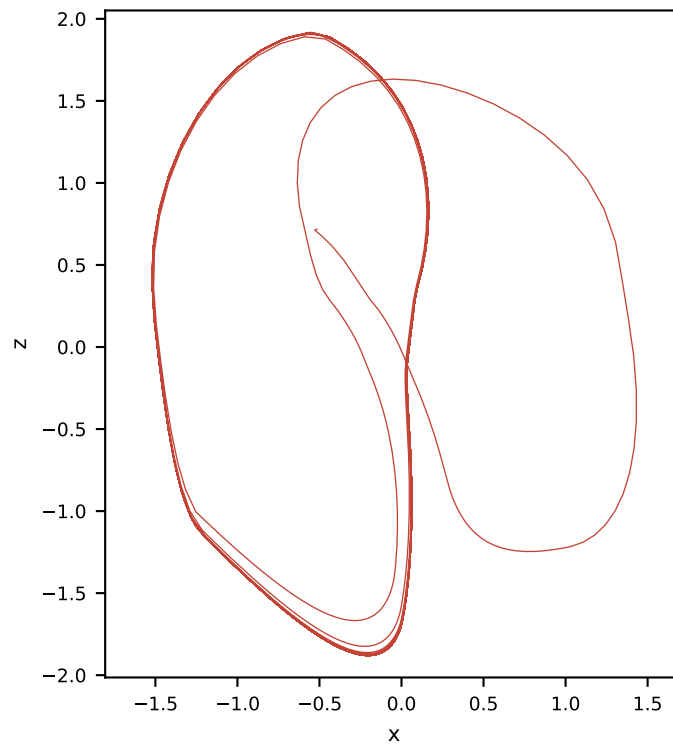
Orthogonal



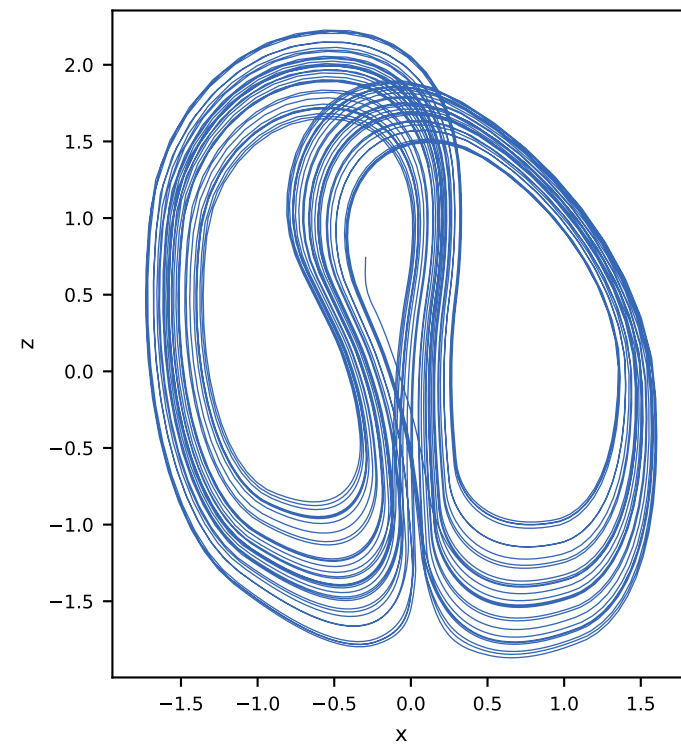
Truth x-z



Vanilla x-z

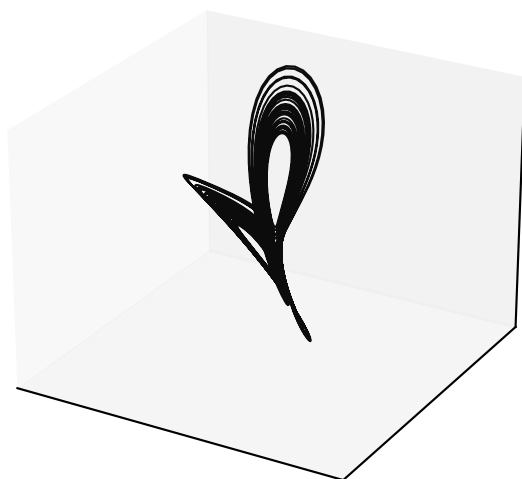


Orthogonal x-z

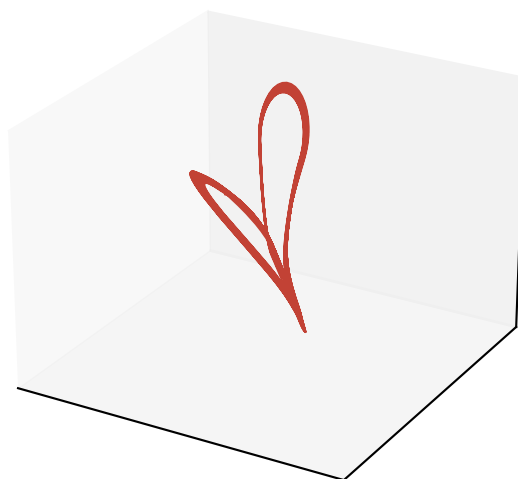


M=9 P=4 (median-PSE seed=2) PSE: van 0.185 ± 0.159 ortho 0.070 ± 0.027 (-62.1%) D_H: van 0.801 ± 0.076 ortho 0.625 ± 0.149 (-22.0%)

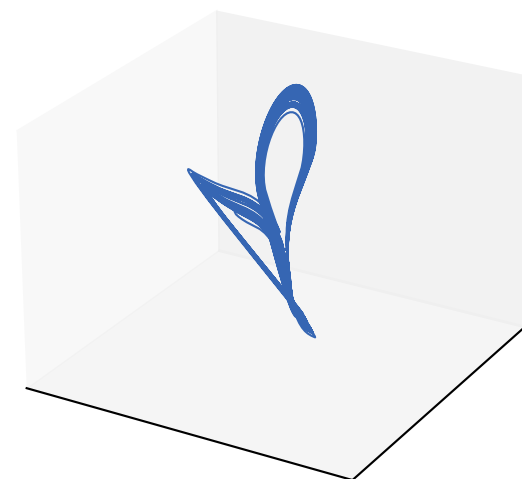
Truth



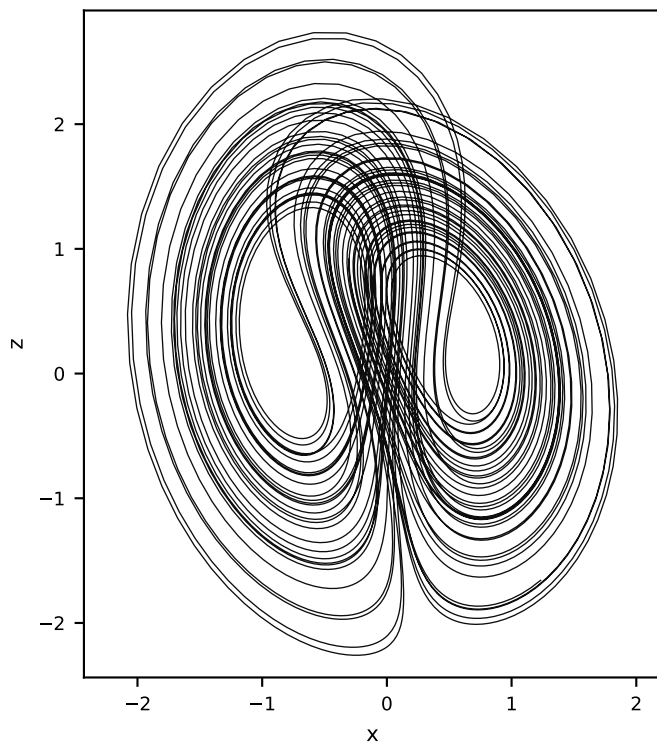
Vanilla



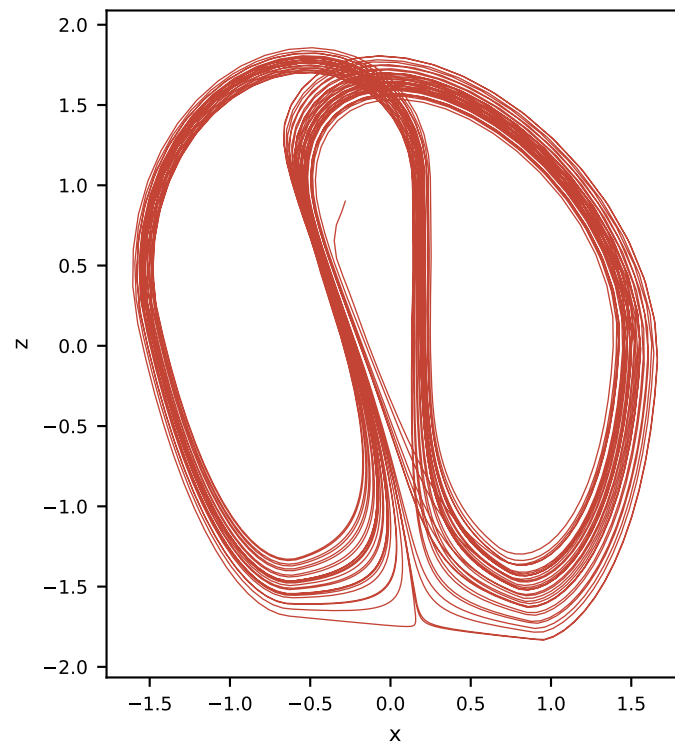
Orthogonal



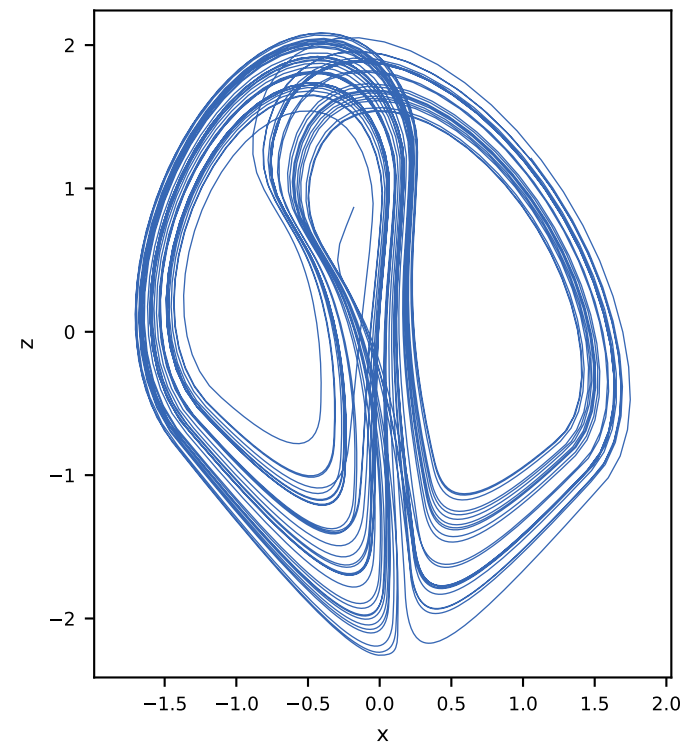
Truth x-z



Vanilla x-z

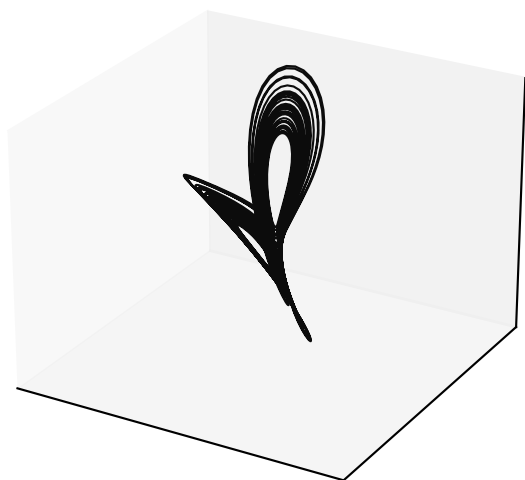


Orthogonal x-z

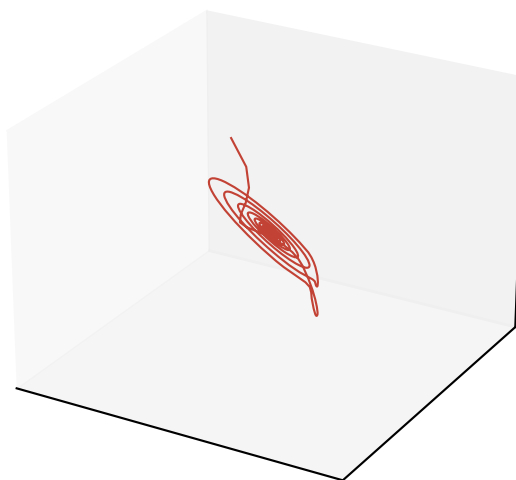


M=5 P=4 (median-PSE seed=2) PSE: van 0.306 ± 0.154 ortho 0.180 ± 0.172 (-41.1%) D_H: van 0.968 ± 0.036 ortho 0.798 ± 0.156 (-17.6%)

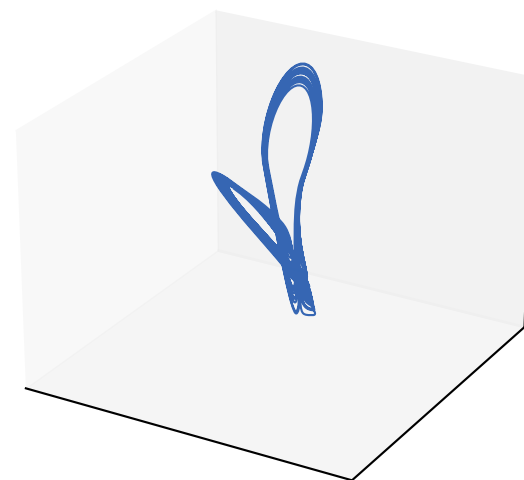
Truth



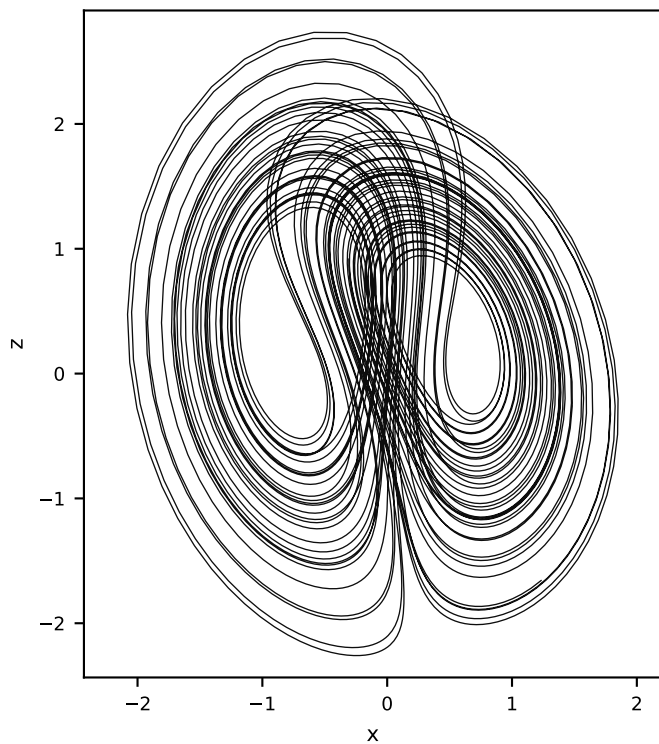
Vanilla



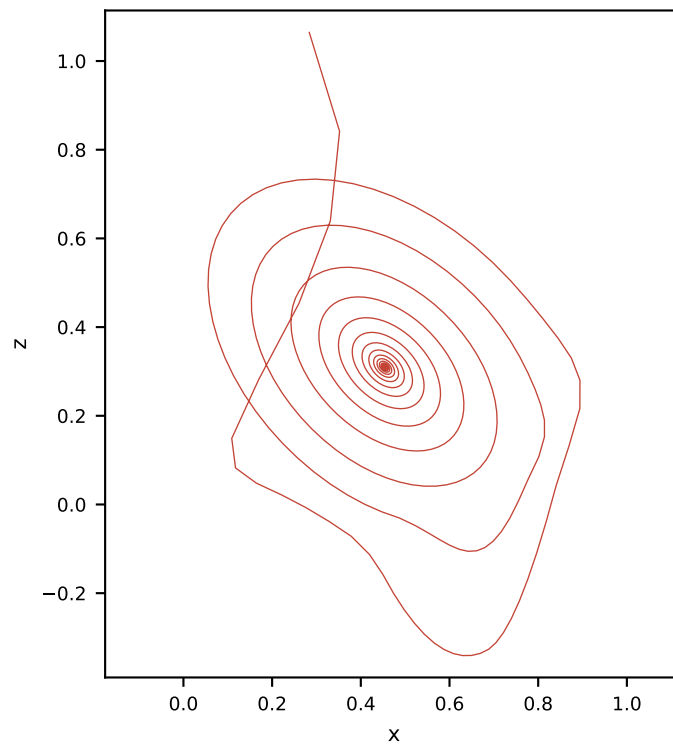
Orthogonal



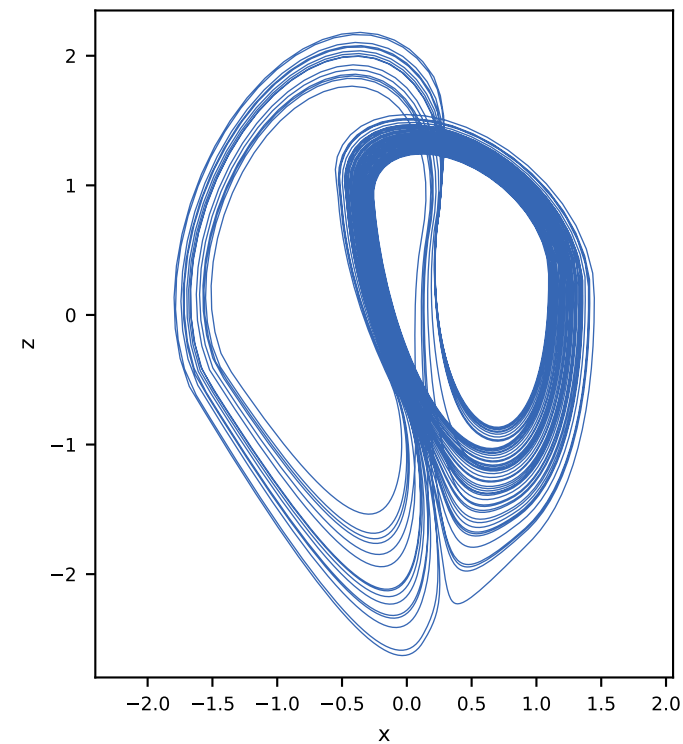
Truth x-z



Vanilla x-z

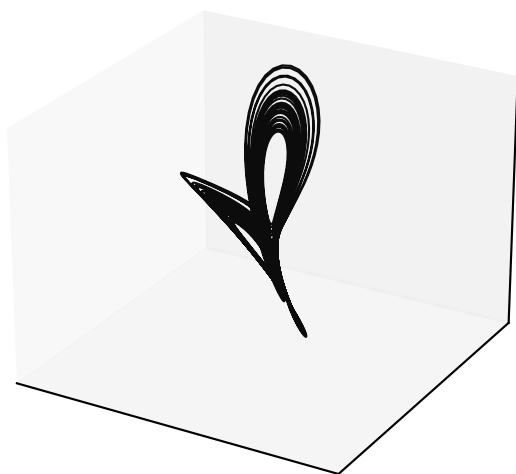


Orthogonal x-z

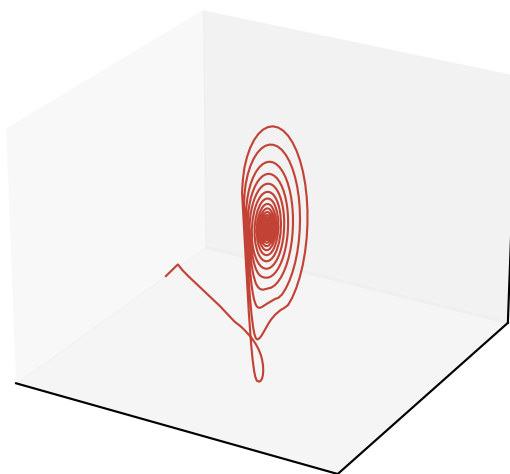


M=5 P=3 (median-PSE seed=0) PSE: van 0.409 ± 0.104 ortho 0.165 ± 0.081 (-59.6%) D_H: van 0.936 ± 0.059 ortho 0.805 ± 0.136 (-14.0%)

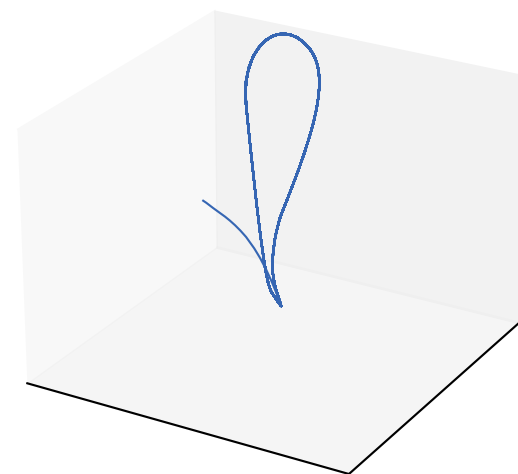
Truth



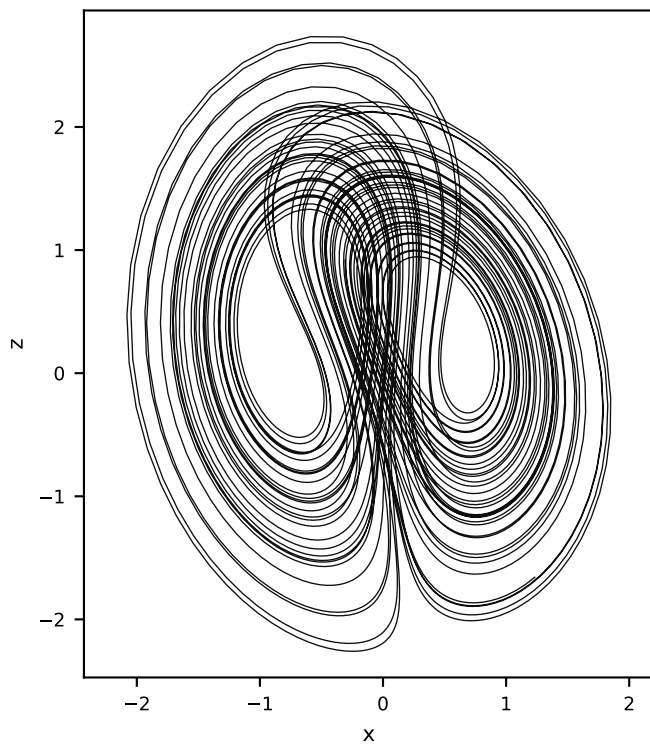
Vanilla



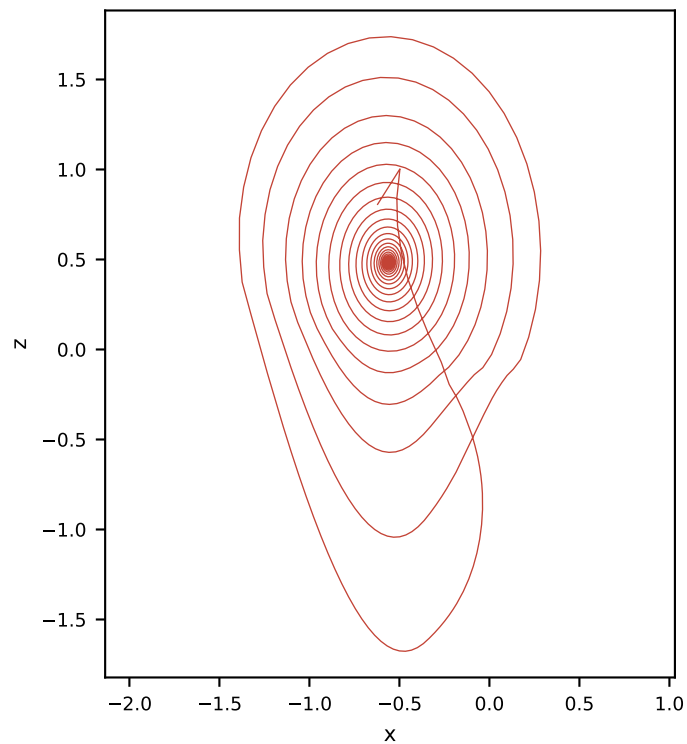
Orthogonal



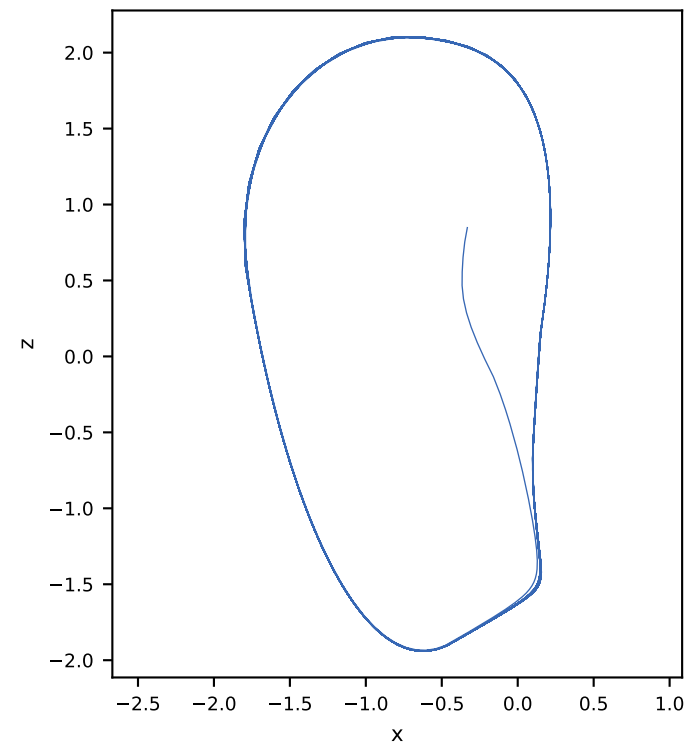
Truth x-z



Vanilla x-z

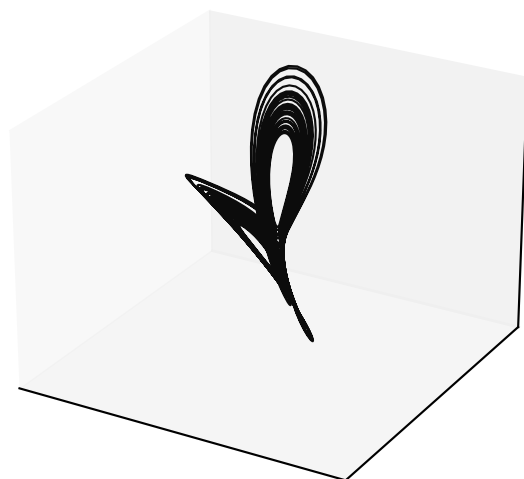


Orthogonal x-z

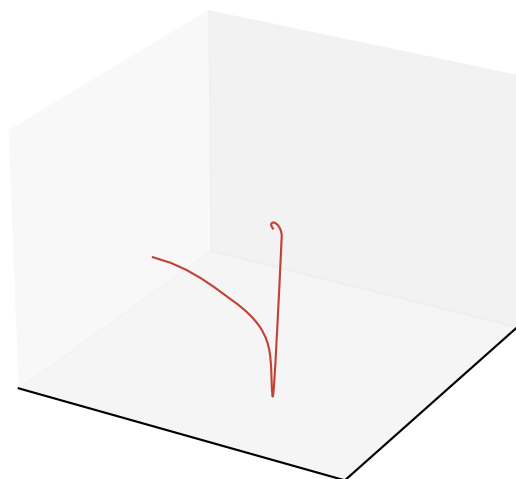


M=4 P=3 (median-PSE seed=1) PSE: van 0.299 ± 0.111 ortho 0.158 ± 0.092 (-47.0%) D_H: van 0.943 ± 0.045 ortho 0.883 ± 0.076 (-6.4%)

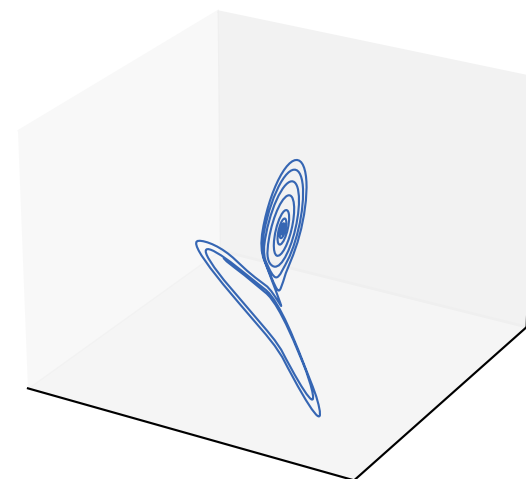
Truth



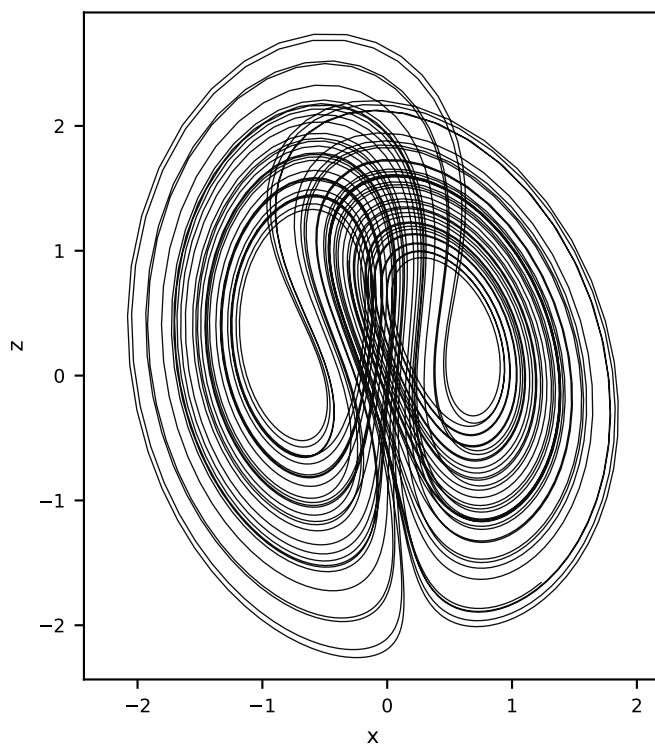
Vanilla



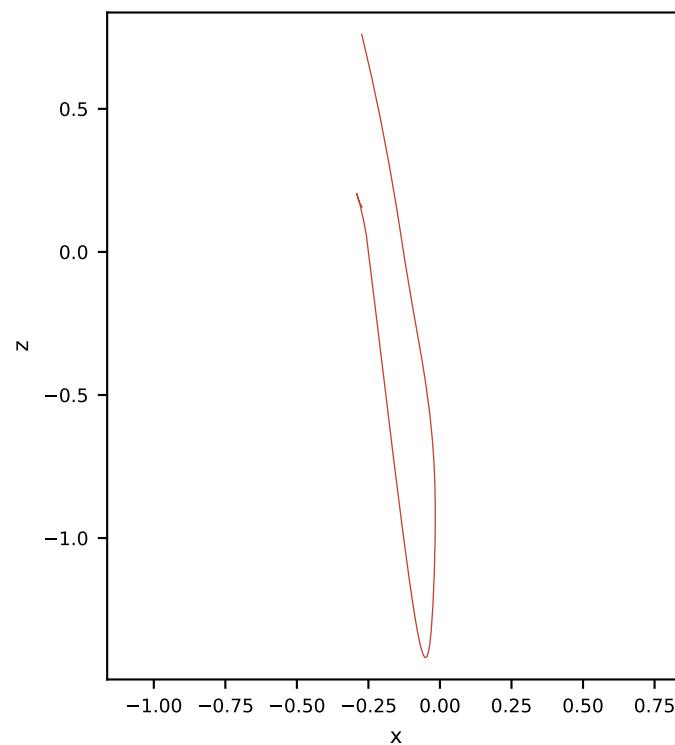
Orthogonal



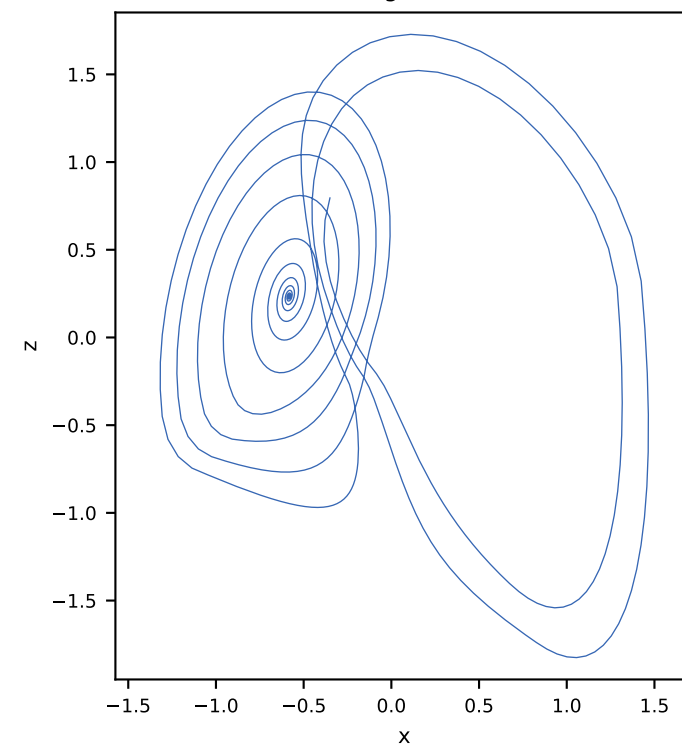
Truth x-z



Vanilla x-z

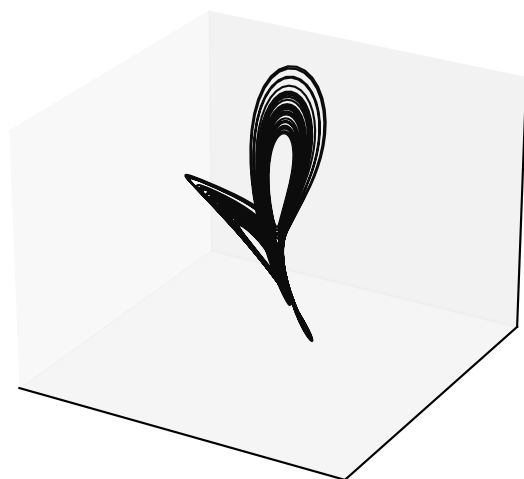


Orthogonal x-z

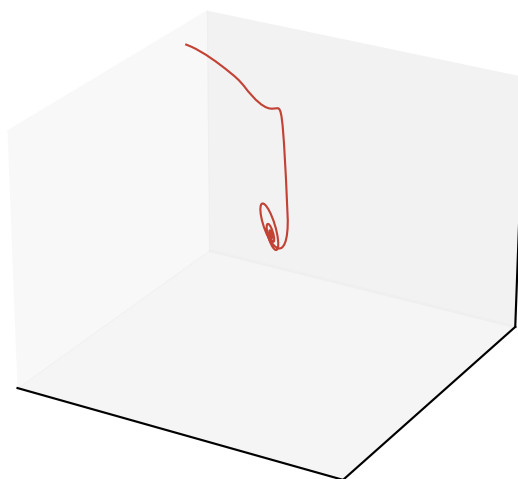


M=4 P=2 (median-PSE seed=0) PSE: van 0.242 ± 0.002 ortho 0.301 ± 0.111 (+24.5%) D_H: van 0.988 ± 0.000 ortho 0.936 ± 0.042 (-5.3%)

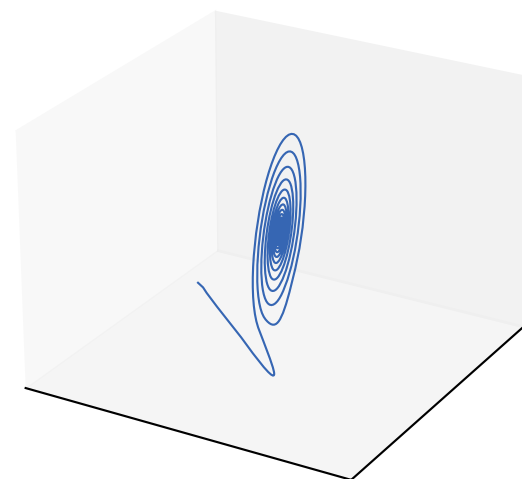
Truth



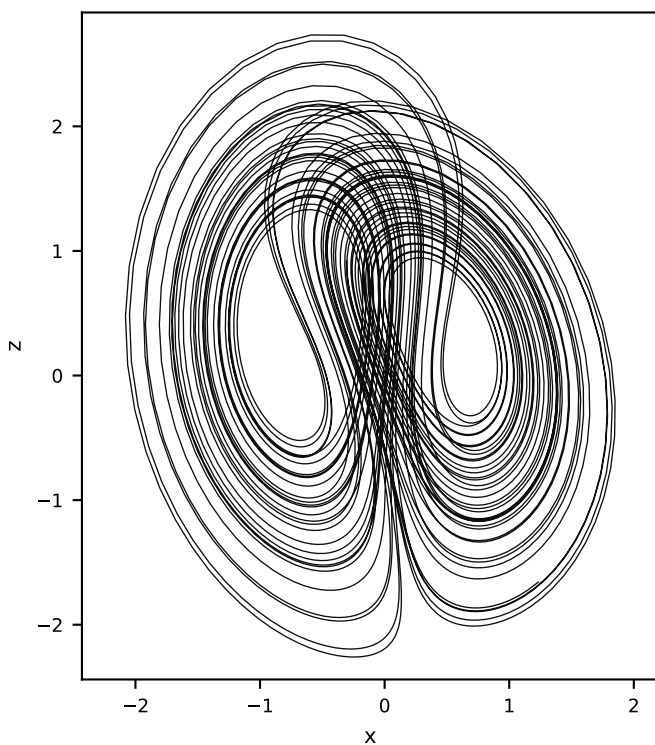
Vanilla



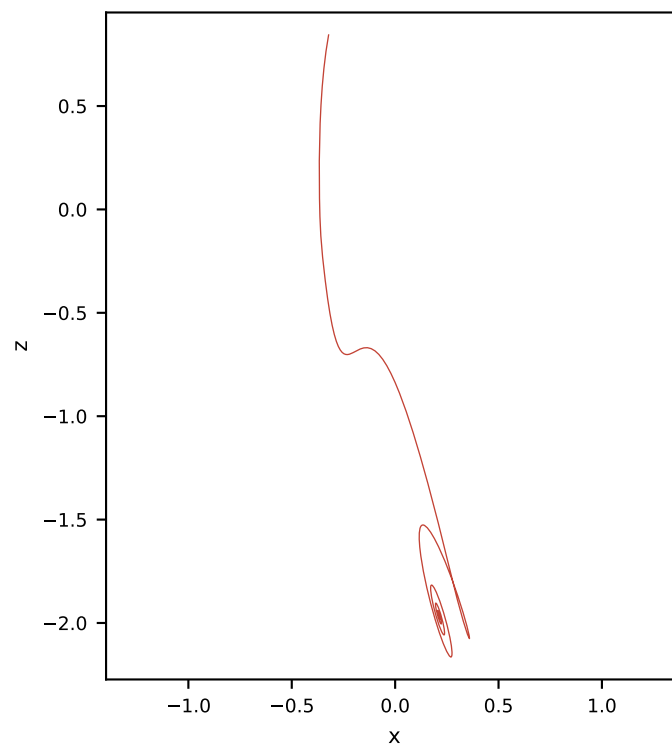
Orthogonal



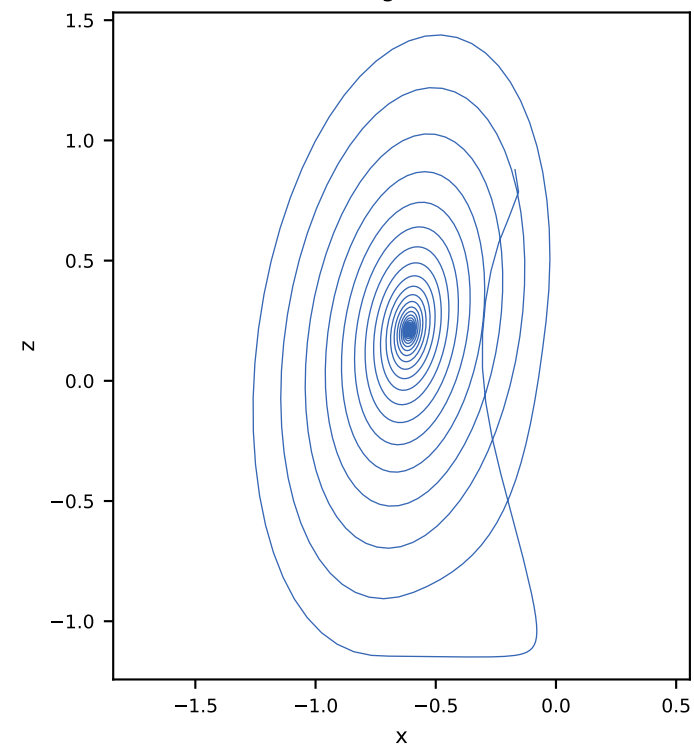
Truth x-z



Vanilla x-z

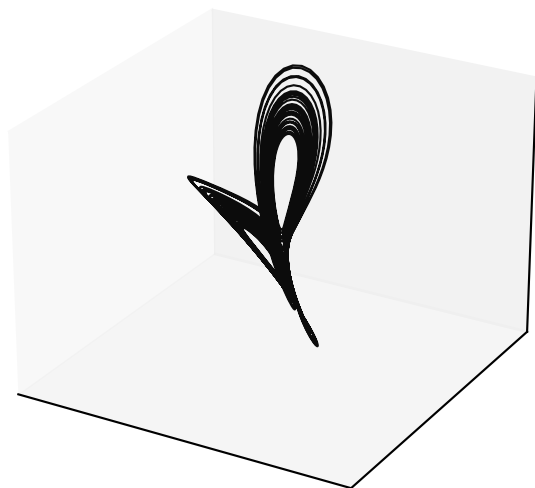


Orthogonal x-z

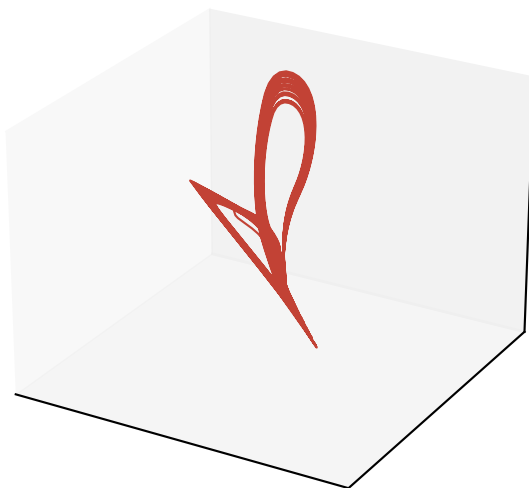


M=9 P=6 (median-PSE seed=1) PSE: van 0.072 ± 0.007 ortho 0.072 ± 0.007 (+0.9%) D_H: van 0.583 ± 0.038 ortho 0.615 ± 0.054 (+5.5%)

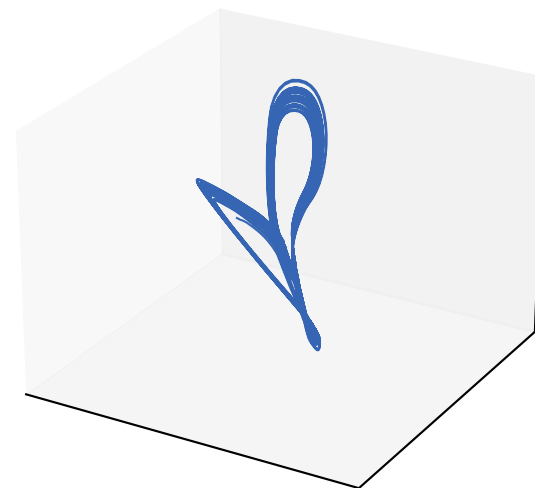
Truth



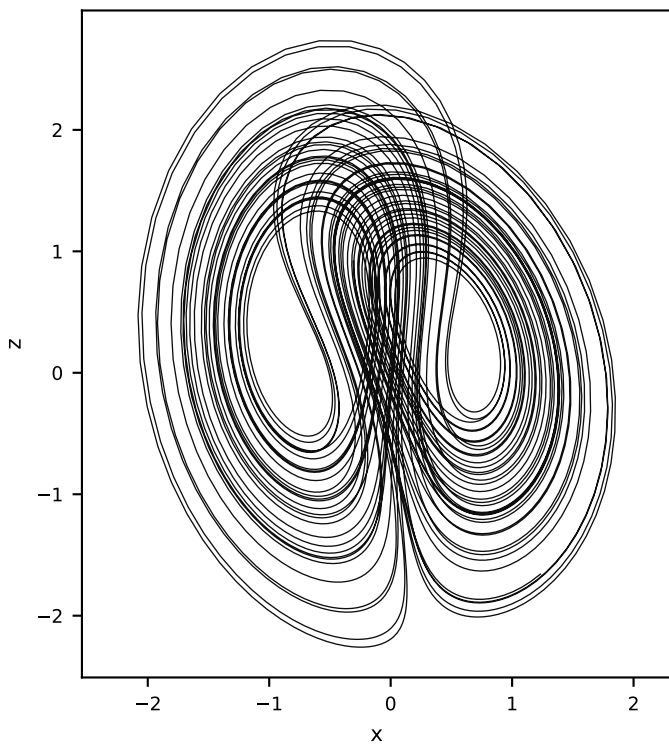
Vanilla



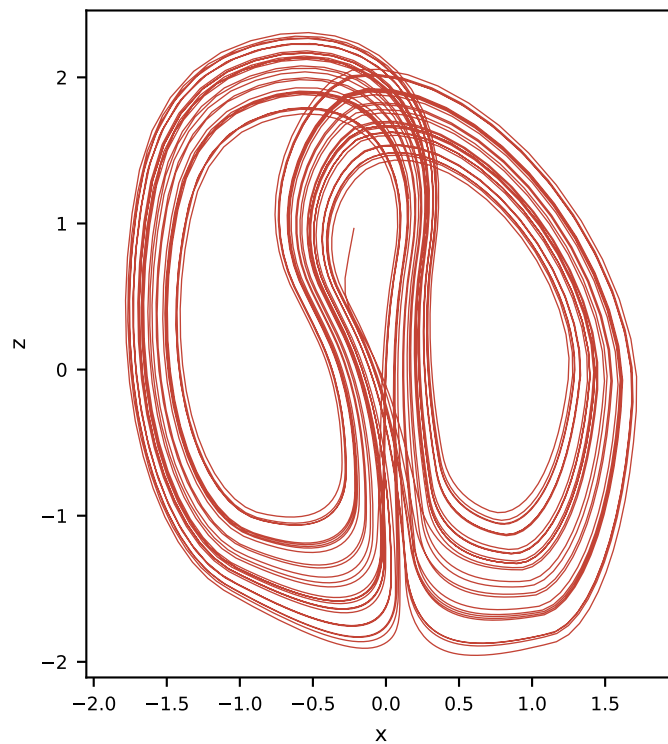
Orthogonal



Truth x-z



Vanilla x-z



Orthogonal x-z

